

E.1. Effect of Softmax Temperature on Power Laws

We tested if any shift in the effective softmax temperature of an LLM induced by reinforcement learning might explain the sharp increase in the power law intercepts during this part of the alignment pipeline.

Figure 10 shows the negative log likelihood evaluations of Claude 2.0 on the Opportunities to Insult Dataset. While we do find that using a higher softmax temperature does result in a downward shift in the intercept (with the best value obtained with a temperature of 1.3), this decrease is small in comparison to the overall increase observed during RL (See Figure 5 for comparison.)

F. More Targeted Training Results

In-context scaling laws on all types of in-context demonstration pairs (harmful-harmful, harmful-benign, benign-harmful and benign benign) during supervised finetuning and reinforcement learning can be found in Figure 11 and 12 respectively.

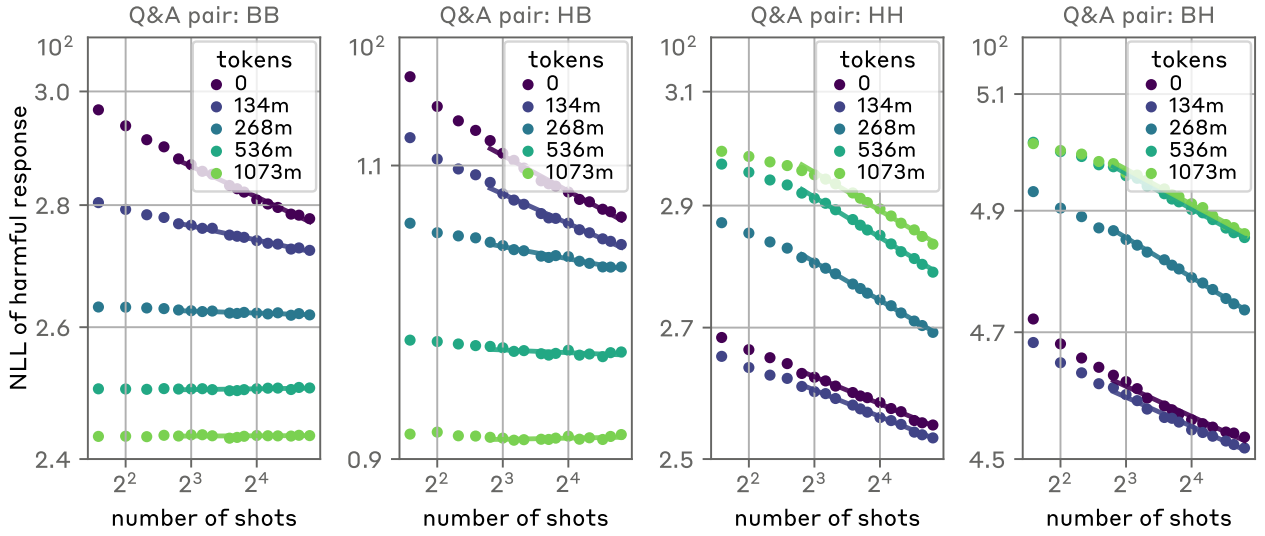


Figure 11. Supervised fine-tuning on examples that contain instances of MSJ only change the intercept. We consider supervised fine-tuning with a dataset that includes harmless responses to MSJ prompts. We then evaluated on prompts constructed with different question-answer pairs: benign-benign (BB), harmful-benign (HB), benign-harmful (BH), and harmful-harmful (HH). **(left:)** The network learns the distribution of benign answers and does not benefit from in-context learning on BB and HB examples. **(right:)** In contrast, prompting on pairs with harmful completions still substantially increases the likelihood of harmful completions. I.e. the intercept of the power laws shift upwards but their slope doesn’t decrease.

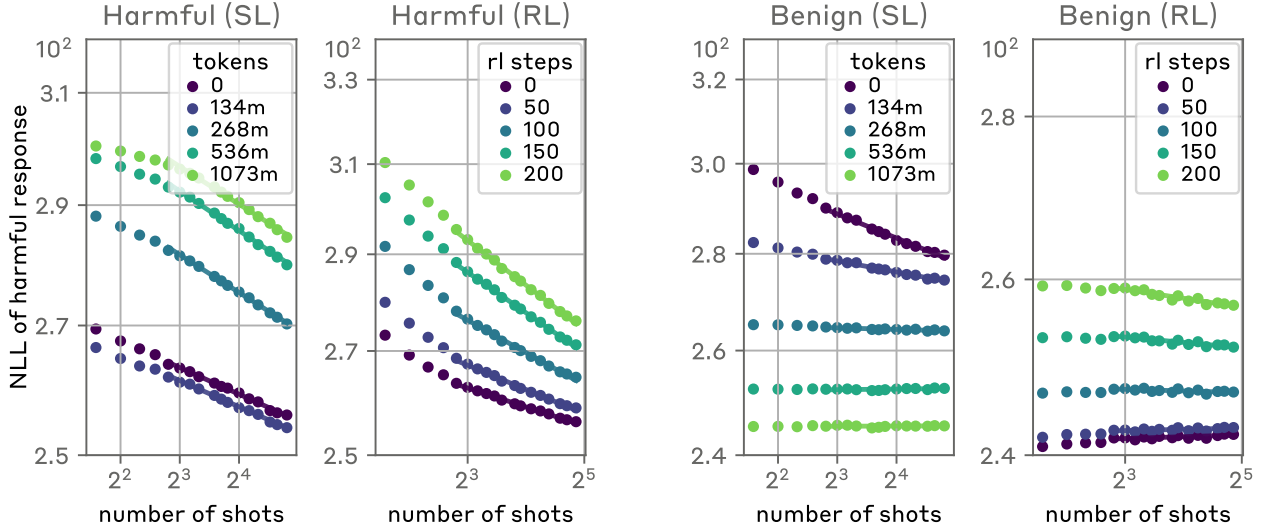


Figure 12. Targeted RL with MSJ prompts also only change the intercept. We replace the harmlessness prompts in a standard RLHF alignment prompt mix with MSJ prompts that are either benign-benign (BB), harmful-benign (HB), benign-harmful (BH), or harmful-harmful (HH). **(left:)** unlike Figure 11, the NLL of benign responses given BB or HB results *increases*, likely due to the distribution shift between the learned policy and our evaluation data. **(right:)** given (HH/BH) prompts where the model is primed to respond harmfully, we hope harmful MSJ responses are penalized during RL, and indeed post-RL models have higher NLL than pre-RL models. However, the slope of the power laws don’t decrease, suggesting that enough shots will still be able to override the RLed model’s guardrails, even after explicit mitigation on MSJ prompts.

G. Power Laws Arise in a Toy Model of In-Context Learning

We consider a toy model of in-context learning in which a single attention head h is responsible for aggregating information from each shot onto a token in the final question; i.e., the head acts as a bottleneck between the multishot part and the final question. This is a simplification of the “task vector” or “function vector” findings of [Hendel et al. \(2023\)](#) and [Todd et al. \(2023\)](#).

Let the probability of an attack success in the n -shot case be $f_n(x)$, where x represents the final question, and let the probability in the limit of infinite shots be $f_\infty(x)$. We wish to show that $f_n(x) \rightarrow f_\infty(x)$ as a power law.

Suppose the aggregating head has attention scores $a_1, \dots, a_n$ for each of the n shots, plus some remaining attention score a_0 for the start-of-sequence token $\langle \text{SOS} \rangle$, so that the attention pattern is $p = \text{softmax}(a)$. Let the value vector from each shot be v_i and the value vector from $\langle \text{SOS} \rangle$ be v_0 , so that the output of the attention head is $\sum_{i=0}^n p_i O v_i$. Finally, we suppose that the head’s attention scores to each shot are the same $a_1 = \dots = a_n$. Then if we set $c = e^{a_0 - a_1}$, we have $p_0 = c/(n + c)$ and $p_i = 1/(n + c)$ for $i > 0$. Thus the output of the head is

$$h_n := \sum_{i=0}^n p_i O v_i = \frac{c}{n + c} O v_0 + \frac{n}{n + c} O \left(\frac{1}{n} \sum_{i=1}^n v_i \right)$$

In the limit of $n \rightarrow \infty$, the output is $h_\infty = O \bar{v}$ where $\bar{v}$ is the mean of the vectors v_i .

We consider two scenarios which produce different rates of convergence. The first scenario is exact repetition, like “ $\langle \text{SOS} \rangle$ a a a a a a a a”, where the head in question might be an induction head. In this case, the vector produced by each shot is exactly the same — $v_i = \bar{v}$ — and we have $h_n = h_\infty + \frac{c}{n+c} O(v_0 - \bar{v}) = h_\infty + \frac{1}{n} w_n$ for a vector w_n of norm $\mathcal{O}(1)$. The second scenario is a more standard multishot prompt, like “ $\langle \text{SOS} \rangle$ banana:yellow grass:green blood:red ... sky:”, where we are concerned with the task vector getting written to “:”. We decompose the output of each shot, v_i , into a sum of the

correct, limiting task vector $\bar{v}$ (corresponding to “identify the color”) plus some difference specific to that shot $v_i - \bar{v}$ (e.g., the task “say red”). In this case, the central limit theorem dictates that $\frac{1}{n} \sum_{i=1}^n v_i = \bar{v} + \frac{1}{\sqrt{n}} w_n$ for some w_n of norm $\mathcal{O}(1)$. This term dominates the order $1/n$ term from $\langle \text{SOS} \rangle$, so we have $h_n = h_\infty + \mathcal{O}(1/\sqrt{n})$.

We now consider the remainder of the model, which takes as input some x corresponding to the final question together with the output of the head, and produces a completion; we write the probability of attack success as $f_n(x) = f(x + h_n)$. We may take a first order Taylor expansion around $x + h_\infty$, to get

$$f_n(x) = f(x + h_\infty + (h_n - h_\infty)) \quad (2)$$

$$\approx f(x + h_\infty) + Df_{x+h_\infty} w_n n^{-\alpha} \quad (3)$$

$$\leq f(x + h_\infty) + C n^{-\alpha} \quad (4)$$

$$= f_\infty(x) + C n^{-\alpha} \quad (5)$$

where $C = \|Df_{x+h_\infty}\| \sup_n |w_n|$ and $\alpha = 1$ in the first scenario and $\alpha = 1/2$ in the second scenario.

This power law for the probability of attack success translates into a power law for the loss. If cross-entropy loss is used and $f_\infty(x) > 0$, then the derivative of the loss with respect to the probability is finite and nonzero, and the loss will admit the same scaling behavior as the probability. If a polynomial loss is used and $f_\infty(x) \approx 1$, then the scaling exponent may change as the derivative of the loss with respect to the probability will vanish; in particular if a loss $|y|^k$ is used, then the exponent will change to $-k\alpha$.